Can a machine curate our playlists?

Our prototype, what we learned, and what we suggest

Our short answer
Yes, as a helper. It makes good first drafts very fast. But it still needs an expert to check the result.

Like sorting mixed balls by colour and size

1
Clean the list
Removed songs listed twice.

2
Make scores comparable
Like converting prices into one currency.

3
Keep what matters
Like the key points of a report.

4
Choose how many
Tested many options, kept the best fit.

5
Sort the songs
Songs that sound alike go together.

Used (9): danceable, energetic, loud, fast, happy, spoken words, acoustic, instrumental, live

Dropped (4): musical key, major or minor, beat count, song length. They don't change how a song feels.

Playlists info (34 playlists, all in range : 50-250 songs)

A good example: Playlist 28

203 calm piano and classical songs: Bach, Beethoven, Chopin, Einaudi. Quiet, no singing. Great for studying or relaxing.

How we checked

Calm classical and heavy metal almost never end up together, and songs by the same artist mostly stay together.

Partly: it hears how a song feels, not its style

Our weak example: one playlist (playlist 23) mixed these songs together

Cheerful pop
Katy Perry
Avicii
ABBA

Same energy, same speed, similar mood
Put in the same playlist

Heavy metal
Metallica
Nirvana
Pantera

What would help: the style of music, the year, the language, and what our users play together and skip.

Good for starting

Pros—very fast, works with millions of songs
Easy to run again
Number of clusters/playlists, we decide

Cons—Each song can be in only one playlist
Choose the number of playlists

Question 5. What would be your next steps if you continued with this project?

Our next steps

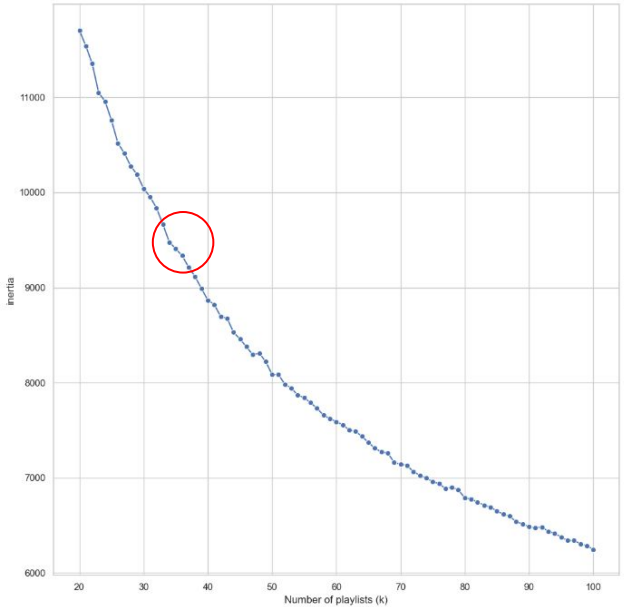
- Listening test
- Fix the weak ones (by tuning number of clusters or scaling)
- Add more information (style, year, language ..)
- Test with customers

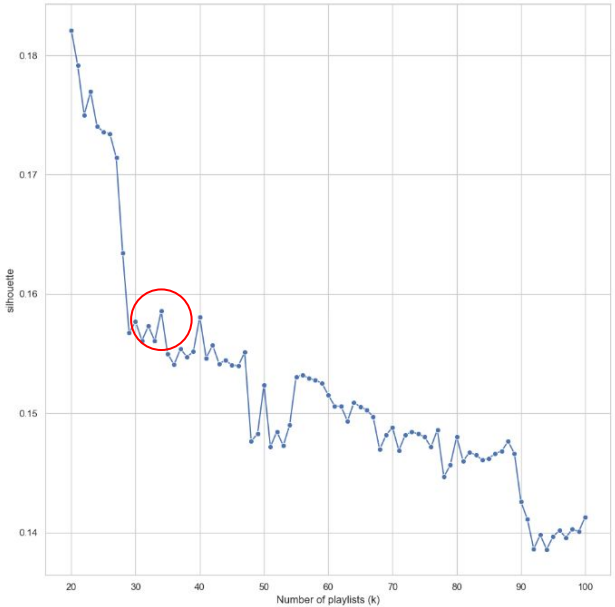
Backup

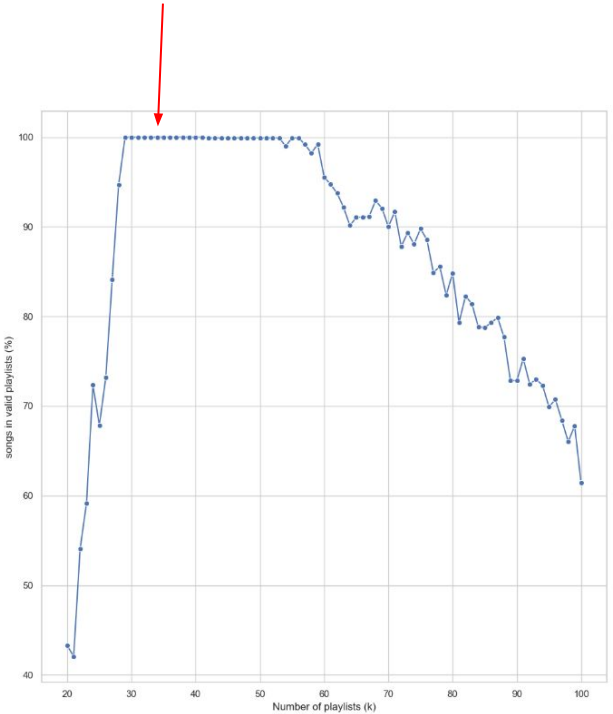
Scaling

	No scaling	MinMax	Robust	Standard	Quantile	Power
danceability	0.0	5.7	3.5	7.7	6.9	7.7
energy	0.0	9.5	5.0	7.7	6.9	7.7
key	0.0	11.9	3.7	7.7	7.6	7.7
loudness	0.0	1.2	11.4	7.7	6.9	7.7
mode	0.0	25.8	2.4	7.7	19.0	7.7
speechiness	0.0	0.7	14.7	7.7	6.9	7.7
acousticness	0.0	14.5	4.2	7.7	6.9	7.7
instrumentalness	0.0	16.1	3.4	7.7	8.9	7.7
liveness	0.0	2.6	11.3	7.7	6.9	7.7
valence	0.0	9.0	3.5	7.7	6.9	7.7
tempo	0.0	2.0	6.0	7.7	6.9	7.7
duration_ms	100.0	0.1	29.1	7.7	6.9	7.7
time_signature	0.0	0.8	1.9	7.7	2.0	7.7

No of Clusters selection:

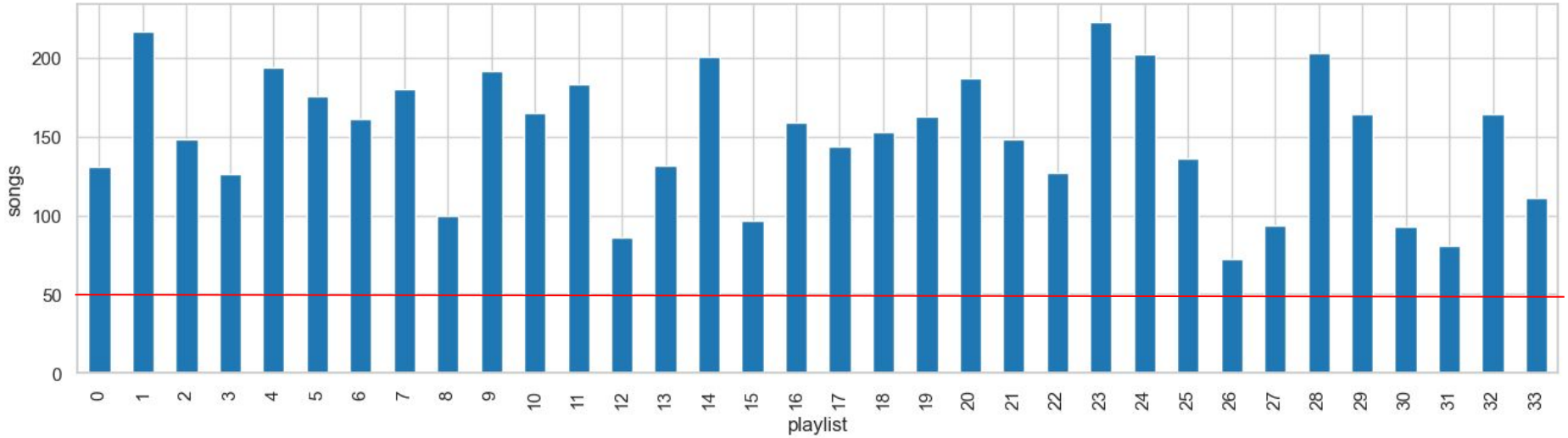






Question 1. How many playlists (clusters) are there?

Playlists info (34 playlists, all in range : 50-250 songs)



Question 2. Is the prototype effective at creating cohesive playlists?

Prototype's performance

Example: Playlist 28 (203 songs)

Calm, acoustic, instrumental, serious mood

Playlist 28

	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo
count	203.000000	203.000000	203.000000	203.000000	203.000000	203.00000	203.000000	203.000000	203.000000
mean	0.186544	0.106071	-24.354724	0.042144	0.921944	0.84330	0.121585	0.069047	77.070675
std	0.081370	0.113851	7.230617	0.006462	0.129589	0.15458	0.075063	0.049207	10.284735
min	0.059600	0.000930	-44.406000	0.027400	0.000580	0.14700	0.056900	0.019600	34.856000
25%	0.124500	0.024000	-28.693500	0.037950	0.916000	0.83450	0.089050	0.036850	70.510000
50%	0.178000	0.070400	-23.785000	0.041700	0.963000	0.90500	0.102000	0.051600	76.591000
75%	0.253000	0.145500	-19.666500	0.045350	0.988000	0.92400	0.119000	0.084100	82.905500
max	0.402000	0.716000	-8.004000	0.070800	0.996000	0.98500	0.762000	0.426000	110.511000

Overall songs

	danceability	energy	loudness	speechiness	acousticness	instrumentalness	liveness	valence	tempo
count	5113.000000	5113.000000	5113.000000	5113.000000	5113.000000	5113.000000	5113.000000	5113.000000	5113.000000
mean	0.506425	0.653386	-9.383185	0.082990	0.291756	0.261381	0.185598	0.441010	118.658878
std	0.217951	0.289807	6.277260	0.070683	0.357076	0.372195	0.149874	0.277642	28.588810
min	0.059600	0.000020	-44.406000	0.022600	0.000000	0.000000	0.010900	0.019600	34.856000
25%	0.304000	0.475000	-11.390000	0.038900	0.002210	0.000001	0.092500	0.195000	96.059000
50%	0.534000	0.738000	-7.371000	0.055400	0.096400	0.002360	0.125000	0.413000	118.988000
75%	0.684000	0.896000	-5.318000	0.098800	0.571000	0.660000	0.237000	0.674000	134.099000
max	0.967000	1.000000	1.342000	0.918000	0.996000	0.985000	0.987000	0.985000	213.990000